Measuring Changes in Pol II Speed & Splicing Time

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Progress Report

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Polymerase II speed

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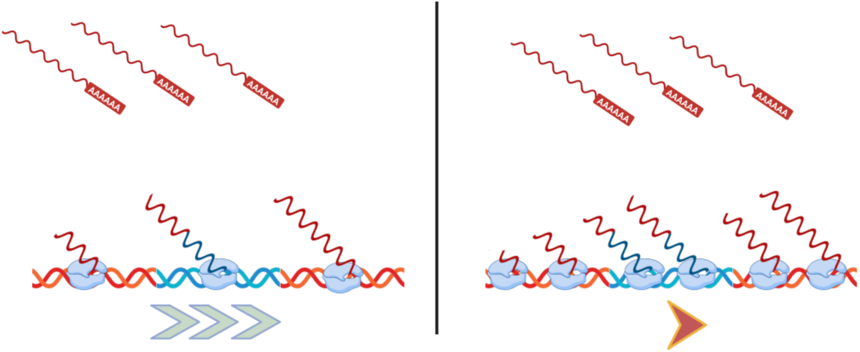
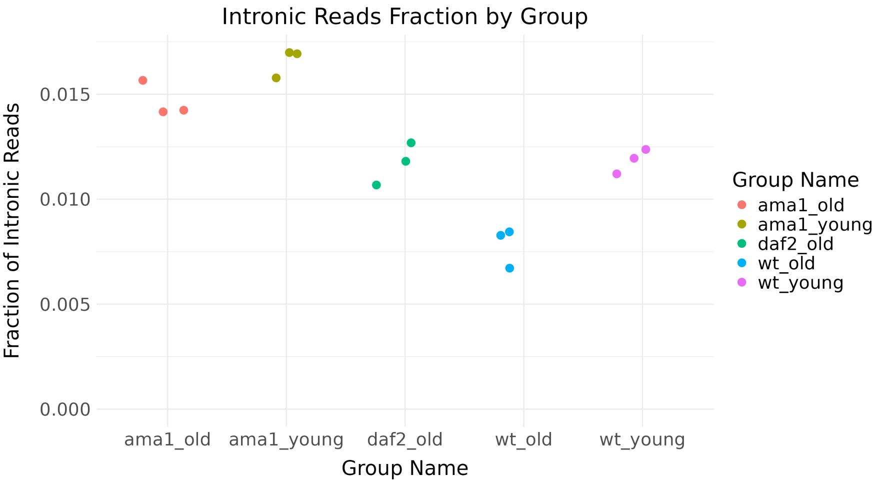
    - E.g. splicing, RNA methylation (deposition of m^6A), alternative poly(A) sites usage, etc.
- We want to estimate log fold changes (LFC) of Pol II speed from total RNA-seq.

Effect of II speed change



- Slower speed $\implies$ higher occupancy of DNA by RNA polymerases $\implies$ more nascent RNA $\implies$ higher fraction of intronic reads.

C. elegans with mutation that slows Pol II



- Total RNA-seq samples of C. elegans with mutation decreasing RNA Pol II speed contain more intronic reads than wild-type control.

The End

Thank you for your attention!

Illustrations and animations were created using BioRender and Manim.